INCH-POUND

MIL-P-14852B (AR)
AMENDMENT 1
22 June 1994

MILITARY SPECIFICATION

PROJECTILE, 105MM: ILLUMINATING,
M314A3, METAL PARTS ASSEMBLY

This amendment forms a part of MIL-P-14852B (AR), dated 30 November 1988, and is approved for use by the U.S. Army Armament, Munitions, and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

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3.4, Delete paragraph in its entirety and substitute with:

"3.4 Rotating band seating.

3.4.1 Rotating band cleanliness. Prior to banding, the band seat shall be clean and free of oil, dirt, grease, rust or other foreign matter. Shot or sand blasting is not permitted for cleaning the band seat after knurling.

3.4.2 Band blank - (gilding metal). The rotating band blank shall be in the form of a solid ring and shall be annealed to comply with the applicable specification. Annealing prior to banding shall not be necessary in the case of hot banding in excess of 800 degrees F.

3.4.3 Application of band blank. The rotating band blank shall not be pressed into the band seat after final heat treatment of the projectile body. The band blank shall be applied in such a manner as not to distort the projectile body.

3.4.4 Sidewall contact. The band blank must contact the sidewall of the band seat for a minimum of eight-tenths of the band seat depth, measured from the outside surface of the projectile body to the rotating band undercut surface.

3.4.5 Sidewall gap. There shall be no gap between the rotating band and the band seat sidewalls at the outside surface of the projectile body after the rotating band has been machined to final dimensions.

3.4.6 Rotating band clearance. The diametrical clearance between the the rotating band and the band seat shall not exceed .006 inch, when the destructive method of testing is used, or .009 inch when the non-destructive method of testing is used."

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3.6, Delete paragraph in its entirety and substitute the following:

"3.6 Proving ground functional tests. The projectile assembly shall reveal no evidence of failure when ballistically tested at excess pressure (see 4.5.5.1). Metal part failure shall be defined as a breakup either in the bore of the weapon or in flight."

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4.4.2.4, Add the following new requirement paragraph to Major category 105:

"3.4.1"

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4.4.2.5: Change Major categoy 104 from "4.4.3.1.2.6" to "4.4.3.1.2.7". Add the following new major defect 106:

"106 Sidewall gap .40% 3.4.5 Gage"

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4.4.3.1.2.7, Delete in its entirety and substitute the following:

"4.4.3.1.2.7 Rotating band. The sample of rotating band for testing in accordance with 4.5.1.2.3 for compliance with 3.3.2 shall be provided as part of the sample taken from 4.4.3.2.1.3. The sample size, frequency and rejection criteria is the same as that described in 4.4.3.2.1.3."

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4.4.3.2.1.3, Delete in its entirety and replace with:

"4.4.3.2.1.3 Sidewall contact. One body assembly shall be selected at the beginning of each shift (not to exceed ten hours) from each shift (not to exceed ten hours) from each banding machine and sectioned in accordance with 4.5.2.3 for compliance with 3.4.4 and also 3.3.2 (see 4.4.3.1.2.7). If, for any reason, there is a change in operator, machinery, or material (by lot number) during a shift, another sample shall be taken. Failure to comply with the requirements of the specification shall be cause for rejection of all material processed from the previous sample."

4.5.1.2.3, Change reference paragraph from "4.4.3.1.2.6" to "4.4.3.1.2.7".

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4.5.2, Delete paragraph in its entirety and replace with:

"4.5.2 Rotating band seating.

4.5.2.1 Clearance (Nondestructive method). The samples provided by 4.4.3.2.1.1 shall be tested for the requirements in 3.4.6 utilizing the equipment andprocedures prescribed in the Detroit Testing Machine Manual (see 6.7).

4.5.2.2 Clearance (Destructive method). The body assembly selected in accordance with 4.4.3.2.1.2 for compliance with 3.4.6 need not be processed through the band machining line, although the band should have a light surface cut (.020" max.) to provide a uniform surface for measurement. The diameter of each band shall be measured at three locations, sixty degrees apart, on the points along the cylindrical portion of their length, near the front and rear edges. This results in six (three pairs of two) diametrical measurements per band. The points of measurement shall be marked on the band and the projectile body adjacent thereto in such a manner as to permit proper body and band orientation after removal. The band shall be carefully removed so as to keep deformation to a minimum. After removal, the thickness of the band shall be measured at each of the marked points (a total of twelve measurements) and the diameter of the band seat corresponding to these points shall also be measured. Add to each of the diameters of the band seat the sum of the two corresponding two thicknesses of the band; the results obtained shall then be subtracted from the previously determined diameters over the band.

4.5.2.3 Sidewall contact. The samples provided in accordance with 4.4.3.2.1.3 shall be sectioned by cutting through two planes normal to the longitudinal axis one-half inch forward of the rotating band. The cylindrical portion containing the rotating band shall be sectioned approximately in half through a plane parallel to its longitudinal axis. One of the two halves shall be used for establishment of mechanical properties of the band; the other section shall be sectioned into two parts, exposing four surfaces for the viewing of the band swaged into the band seat. The exposed surfaces shall be ground, and the metal chip shall be removed prior to measurement for depth of sidewall contact to determine compliance with 3.4.4."

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4.5.5.1, Change the second sentence to the following:

"A propellant charge adjusted to yield a mean pressure of 42,500 psi with no maximum individual exceeding 44,400 psi shall be used with the applicable projectile."

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Change the last sentence of 4.5.5.1 to the following:

"Recovery and examination of metal parts for any abnormal condition shall be accomplished for information only."

Custodian:
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